Scheduled Transit as Deliberate Inversion

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Abstract

Scheduled transit is not a failed application of the consolidation model; it is a deliberate inversion of it. Where the matatu fixes the load factor and lets departure time float, scheduled transit fixes the departure time ex ante and lets the load factor become the residual. This note derives the subsidy as the price of the schedule, shows that the consolidation frontier is violated by design, catalogues the three pathologies that follow (no kill rule, no yield management, no capacity adaptation), and characterises the matatu as the spot market and the bus as the forward contract. The design implication is a corridor-density rule: scheduled service where density clears the threshold, a trunk plus feeder at medium density, and demand-responsive service at low density, where waiting is the only consolidation mechanism.

Yes — and the theory explains it with a precision that should make transit planners uncomfortable. Scheduled transit is not a failed application of the consolidation model. It is a deliberate inversion of it, and the losses are the price of the inversion.

1 The Core Inversion: The Schedule Fixes Time, Load Factor Becomes the Residual

In the consolidation model, departure time is the decision variable and load factor is the output. You wait until the vehicle crosses λ^*, then you leave. Waiting is the consolidation mechanism.

Scheduled transit reverses this. The departure time is fixed ex ante—published, committed, legally binding—and the load factor becomes whatever the market delivers at that time and place.

The moment the timetable is published, the operator has paid c_0 in advance and forfeited the right to recover it by waiting. A bus that must leave at 7:15am leaves at 7:15am whether it carries 60 passengers or 6. The temporal consolidation lever has been amputated by design.

This is why the matatu waits and the bus doesn't. The matatu is solving the consolidation problem. The bus has been forbidden from solving it.

2 The Subsidy Is the Price of the Schedule

Formally, the loss on a scheduled route is:

$$\text{Subsidy}_r = c_0 \left(1 - \frac{\lambda_r}{\lambda^*}\right) + \text{coverage premium}, \tag{1}$$

where the first term is the load-factor gap and the second is the cost of running where density cannot support any schedule at all.

The subsidy is not inefficiency. It is the purchase of a public good: predictability. A passenger who cannot predict when the bus arrives cannot plan a job interview, a school pickup, or a hospital visit. The schedule is the product, and the subsidy is its price.

But this creates a specific pathology: because the operator does not bear the load-factor risk, they have no incentive to consolidate. The subsidy removes the discipline that forces the matatu to wait. The result is that scheduled systems systematically operate at lower load factors than informal ones—not because they are worse managed, but because the management objective has been changed from "fill the vehicle" to "run the timetable."

3 The Frontier Is Violated by Design, Not by Accident

The consolidation frontier says: higher load factor requires higher consolidation cost (waiting, circuity, acquisition). Scheduled transit chooses to minimise consolidation cost—no waiting, no circuity, fixed routes—and accepts whatever load factor results.

This is a coherent choice. It is the choice to be on the low-consolidation, low-load-factor end of the frontier. The matatu is on the high-consolidation, high-load-factor end. Neither is wrong in the abstract. They are optimising different objectives: the matatu optimises contribution per departure; the bus optimises coverage per dollar of subsidy.

The loss is the difference between the two objectives, and it is real. It is also the thing the public is buying.

4 Three Pathologies That Follow

1. **No kill rule.** Scheduled transit runs the 2am bus with three passengers because the timetable says so. The matatu does not run at 2am because no one is there. The kill rule—the mechanism that prevents dispatching below λ^*—is exactly what the public-service obligation forbids. You cannot kill the 2am bus if the nurse needs to get home. This is the single largest source of structural loss in scheduled systems, and it is a feature, not a bug.

2. **No yield management.** Airlines fill seats by price discrimination: advance purchase discounts, Saturday-night stays, premium cabins. Scheduled transit usually has flat fares for equity and political reasons. This means it cannot use price to shift demand from peak to off-peak, cannot fill empty off-peak vehicles with discounted fares, and cannot charge the peak passenger the marginal cost of the peak. The yield-management lever is unavailable, so the off-peak runs empty and the peak runs overcrowded.

3. **No capacity adaptation.** The matatu industry discovered the baby matatu—smaller vehicles that fill faster in low-density corridors. Scheduled transit usually cannot do this: the fleet is fixed by procurement cycles, labour agreements, and depot infrastructure. The capacity choice K^* is locked in for a decade, while demand shifts annually. The result is a fleet that is too large for the off-peak and too small for the peak, and cannot be resized.

5 The Matatu Is the Spot Market; the Bus Is the Forward Contract

This is the cleanest way to state the difference.

The matatu sells a departure when it is full. The passenger buys a seat on a vehicle that leaves when the load factor is sufficient. The price is low because the operator bears no risk of running empty—they simply wait.

Scheduled transit sells a forward contract on a departure at a fixed time. The operator commits to leaving whether full or not. The passenger buys certainty. The price is higher—or, if subsidised, the subsidy is higher—because the operator bears the risk of running empty.

Forward contracts always cost more than spot purchases, because the seller bears the risk. The subsidy is the risk premium. This is not a market failure. It is the price of certainty.

6 The Design Implication the Theory Gives

The theory does not say scheduled transit is wrong. It says scheduled transit is viable only where density supports the schedule, and the subsidy required is a function of the density gap.

This yields a clean design rule:

Corridor density	Correct mode	Why
High density (many passengers per hour)	Scheduled service, flat fare, high frequency	λ clears λ^* without waiting; schedule is nearly free
Medium density	Scheduled trunk + demand-responsive feeder	Trunk carries the load factor; feeder absorbs the variance
Low density	Demand-responsive only	No schedule can fill; waiting is the only consolidation mechanism, so let the vehicle wait

This is exactly what cities are discovering empirically—microtransit, on-demand shuttles, flexible route services—and the theory explains why it works. Demand-responsive transit restores the temporal consolidation lever that the schedule had amputated. The vehicle waits (or routes dynamically) until the load factor is sufficient, and the passenger trades certainty for availability.

The matatu is a demand-responsive system with no schedule, no subsidy, and no coverage mandate. It is efficient at load factor and terrible at coverage. Scheduled transit is the reverse. The correct system is a portfolio of both, allocated by corridor density—and the theory tells you exactly where the boundary lies.

7 The Scheduled-Transit Death Spiral

There is a vicious cycle in scheduled transit that is the flywheel running backwards.

As ridership falls, the subsidy per passenger rises, which creates political pressure to cut service, which makes the service less useful, which cuts ridership further. This is the same death spiral as the logistics corridor, but operating through the coverage mandate rather than through the price mechanism.

The difference from the matatu is that the matatu can shrink capacity—smaller vehicles, fewer trips, exit the corridor entirely. Scheduled transit cannot exit, because the coverage mandate forbids it. So the spiral is not self-correcting. It is corrected only by subsidy increases, which are politically costly, or by service redesign, which is administratively slow.

This is why transit agencies in the US and Europe are perpetually in crisis. They are running a flywheel that cannot turn in the virtuous direction, because the kill rule that would restore load factor is exactly the thing the public good forbids.

8 The One-Line Answer

Scheduled transit is loss-making by design: it amputates the temporal consolidation lever that informal systems rely on, fixes the departure time as a commitment rather than a decision, and accepts whatever load factor results—and the subsidy is the price of the predictability that the schedule provides. The matatu is the spot market; the bus is the forward contract; and the forward contract costs more because the seller bears the risk of running empty.

9 Conclusion

The theory's contribution here is not to condemn scheduled transit. It is to say precisely what the subsidy is buying (coverage and predictability), where it is structurally necessary (low-density corridors), and where it is pathological (high-density corridors running below λ^* because no one is allowed to consolidate). That distinction is the difference between a transit agency that defends its subsidy and one that redesigns its network.